

TRACE: Trajectory Deformation for Full-Body Reactive Obstacle Avoidance

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Abstract—Learning from Demonstration (LfD) has emerged as an important paradigm for enabling robots to perform complex downstream manipulation tasks. However, demonstrations are often collected in obstacle-free environments, while real-world execution may involve previously unseen obstacles appearing along the robot trajectory. Although there are several approaches to avoid obstacles, they suffer from important limitations. First, many existing methods focus primarily on the safety of the end-effector while neglecting collisions involving intermediate robot links. Second, locally reactive avoidance strategies often fail to preserve global smooth and kinematically consistent motion. Third, real-time obstacle avoidance in cluttered environments remains computationally challenging due to the need to jointly handle multiple constraints and full-body robot interactions. To address these challenges, we propose TRACE, an RMP-based full-body obstacle avoidance framework that deforms a demonstrated trajectory while remaining as close as possible to the original motion. Our approach generates smooth, collision-free trajectories that avoid both end-effector and intermediate-link collisions through configuration-space Riemannian Motion Policies (RMPs). Experimental results demonstrate that the proposed method preserves the semantic structure of the demonstrated motion while achieving effective obstacle avoidance of the entire body.

Index Terms—obstacle avoidance, Riemannian motion policies, configuration space, policy transportation, robot manipulation, full-body avoidance

I. INTRODUCTION

Trajectory generation methods such as Learning from Demonstration (LfD) [1]–[3] and motion planning provide robots with feasible motions for complex manipulation tasks. In many practical settings, however, trajectories are generated in controlled or partially known environments and later executed in scenes containing previously unseen obstacles. A trajectory that was initially valid can therefore become unsafe during deployment, colliding with surrounding objects, furniture, or even near humans. The central challenge is to adapt nominal trajectories online to new environments without requiring costly replanning or re-demonstration, while simultaneously preserving the smoothness, structure, and semantic intent of the original motion.

Existing approaches address this challenge from different angles. Trajectory optimisation and reactive motion policies [4]–[8] can produce collision-free motion in real time, but each family of methods carries its own set of limitations. Optimisation-based methods typically assume a known envi-

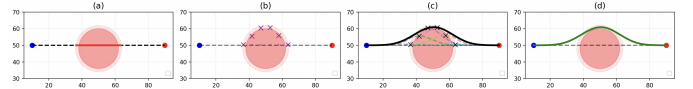


Fig. 1. TRACE EEF Obstacle avoidance pipeline: (a) shows that in the path there is obstacle present (b) shows the gaussian means are initialized at the surface (c) shows a set of gaussian with QP bump and (d) shows the final output

ronment model and are expensive to rerun online. Reactive controllers respond to obstacles locally, but can produce jittery, task-inconsistent corrections. Critically, the vast majority of previous work targets only the End-Effector (EEF): the arm’s elbow and intermediate links are left unmonitored, so the robot can dodge an obstacle with its hand while simultaneously hitting it with its forearm.

We propose Trajectory Deformation for Full-Body Reactive Obstacle Avoidance (TRACE), a two-stage full-body avoidance framework. In the first stage, a geometric refinement pipeline deforms the demonstrated trajectory in Cartesian space, pushing the EEF path away from any detected obstacles while keeping the deformed path as close as possible to the original. In the second stage, a Configuration-space Riemannian Motion Policy (C-space Riemannian Motion Policy (RMP)) layer monitors every robot link online and generates coordinated joint corrections that prevent any part of the arm from entering a collision volume. Together, the two stages complement each other: the offline deformation resolves straightforward collisions efficiently and analytically, while the online RMP layer provides rigorous, real-time full-body guarantees for more complex or dynamic scenarios. Figure 1 illustrates how the EEF trajectory has changed.

II. GEOMETRIC TRAJECTORY REFINEMENT

Given a trajectory warped by a policy transportation method [3], some waypoints may still penetrate obstacles present in the deployment scene. This section describes how TRACE resolves those collisions offline, producing a smooth, collision-free Cartesian path before any joint motion is commanded.

a) Collision Region Proposal: The simplest fix is to detect every waypoint that lies inside an obstacle and push it out by projecting it onto the obstacle surface. One could then interpolate the displaced points with a Gaussian process (GP) to recover continuity. This strategy fails in two ways.

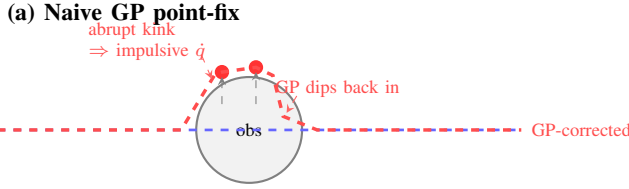


Fig. 2. Why a naive GP point-fix fails and how the principled approach resolves it. (a) Projecting colliding waypoints onto the obstacle surface and fitting a GP introduces abrupt kinks in the path (large impulsive joint velocities) and offers no hard safety guarantee—the GP can dip back inside the obstacle between corrected points.

First, GP regression introduces smooth but uncontrolled errors: even with zero residual at the corrected points, the GP can dip back inside the obstacle between them, offering no hard safety guarantee as shown in Figure 2. Second, isolated point corrections create abrupt direction changes in the path, which translate into large, impulsive joint velocities when the trajectory is tracked on hardware.

A more principled solution must (i) detect collisions along *segments*, not just at discrete points; (ii) generate displacements that are smooth *by construction*; and (iii) keep the modified path as close as possible to the original. The following subsections develop exactly this.

Discrete point-wise collision checks may miss collisions that occur between sampled waypoints. TRACE therefore treats each pair of consecutive waypoints as a line segment and computes the minimum clearance of that segment to every obstacle.

For a segment from $\vec{a}$ to $\vec{b}$ and an obstacle centred at $\vec{c}$ with radius r_{obs} , the closest point on the segment is found via orthogonal projection:

$$u = \frac{(\vec{c} - \vec{a}) \cdot (\vec{b} - \vec{a})}{\|\vec{b} - \vec{a}\|^2}, \quad u^* = \max(0, \min(1, u)). \quad (1)$$

The minimum clearance of the segment to the obstacle centre is

$$d = \|\vec{c} - (\vec{a} + u^*(\vec{b} - \vec{a}))\|, \quad (2)$$

and a continuous collision weight that blends smoothly from zero (no collision risk) to one (full penetration) is defined as

$$\alpha = \max\left(0, \min\left(1, \frac{r_{\text{ref}} + r_{\text{obs}} + \epsilon - d}{\epsilon}\right)\right). \quad (3)$$

This weight will later gate the magnitude of the evasive displacement, avoiding abrupt corrections near obstacle boundaries.

b) Localized Evasion Frame: Once a segment is flagged as colliding, we need to decide *in which direction* to push the trajectory. Perturbing a waypoint along an arbitrary Cartesian axis risks distorting the arc length profile and the curvature of the path. The natural choice is to restrict evasive motion to the plane orthogonal to the trajectory tangent, since displacements in that plane leave the path length and the along-track progression unchanged.

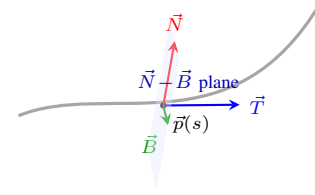


Fig. 3. Localized Frenet-Serret frame at $\vec{p}(s)$. Evasive search is restricted to the $\vec{N}$ - $\vec{B}$ plane to prevent trajectory self-intersection.

The unit tangent at each waypoint is

$$\vec{T} = \frac{\dot{\vec{p}}}{\|\dot{\vec{p}}\|}. \quad (4)$$

Using a fixed reference vector $\vec{r} = [0, 0, 1]^\top$, a local Frenet-Serret-like frame is constructed:

$$\vec{N} = \frac{\vec{r} - (\vec{r} \cdot \vec{T})\vec{T}}{\|\vec{r} - (\vec{r} \cdot \vec{T})\vec{T}\|}, \quad \vec{B} = \vec{T} \times \vec{N}. \quad (5)$$

All evasive displacements are expressed as linear combinations of $\vec{N}$ and $\vec{B}$, guaranteeing that they are orthogonal to the nominal trajectory as shown in Figure 3

c) Archimedean Spiral Search: Within the $(\vec{N}, \vec{B})$ plane we must find the nearest point that clears the obstacle. A brute-force grid search over this plane is wasteful because collision-free points typically lie close to the obstacle boundary. Instead, TRACE samples candidate evasive points along an Archimedean spiral (Figure 4) that starts at the obstacle boundary and expands outward:

$$\vec{p}_{\text{eval}}(\theta) = \vec{p}_{\text{path}} + \frac{b}{2\pi}\theta \cos \theta \vec{N} + \frac{b}{2\pi}\theta \sin \theta \vec{B}. \quad (6)$$

The search terminates as soon as a candidate point achieves the required clearance, keeping the displacement as small as possible and therefore as close as possible to the original trajectory.

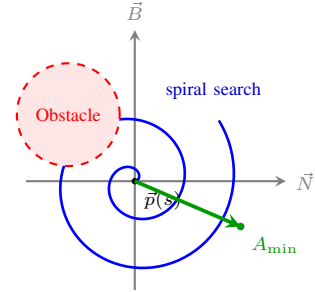


Fig. 4. Archimedean spiral search radiating from $\vec{p}(s)$ in the $\vec{N}$ - $\vec{B}$ plane until minimum clearance A_{min} is found.

d) Smooth Trajectory Refinement via QP: Applying the spiral-found displacement independently at each colliding waypoint would create a piecewise-corrected path with sharp kinks where corrected and uncorrected segments meet. To obtain globally smooth evasions, TRACE couples the displacements across all waypoints through a quadratic programme (QP):

$$\min_{\mathbf{u}} \quad \frac{1}{2} \mathbf{u}^\top \mathbf{I} \mathbf{u} \quad (7)$$

$$\text{s.t.} \quad \mathbf{M} \mathbf{u} \geq \mathbf{A}_{\text{min}}. \quad (8)$$

The constraint matrix M encodes Gaussian influence weights

$$W(s_k, s_j) = \exp\left(-\frac{(s_k - s_j)^2}{2\sigma^2}\right), \quad (9)$$

so that the displacement solved for at one waypoint spreads smoothly to its neighbours (Figure 5). The QP minimises the total displacement magnitude subject to every colliding waypoint receiving at least the clearance prescribed by the spiral search, producing a smooth bump in the trajectory that fades gracefully back to the original path on either side of the obstacle as given in Figure 5

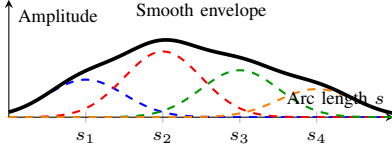


Fig. 5. QP bump: discrete evasion amplitudes (dashed) at colliding waypoints spread via Gaussian spatial kernels, producing a smooth, continuous deformation along the trajectory arc length.

e) Joint-Space Projection and Null-Space Refinement:

The Cartesian evasive displacement $\Delta \vec{x}_{\text{eef}}$ is converted into joint motion using a damped least-squares (DLS) inverse Jacobian:

$$\mathbf{J}^\dagger = \mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1}. \quad (10)$$

The resulting joint update is

$$\Delta \mathbf{q} = \underbrace{\mathbf{J}^\dagger \Delta \vec{x}_{\text{eef}}}_{\text{Primary task}} + \underbrace{(\mathbf{I} - \mathbf{J}^\dagger \mathbf{J}) (-k_p (\mathbf{q} - \mathbf{q}_{\text{pref}}))}_{\text{Null-space posture term}}. \quad (11)$$

The first term performs EEF obstacle avoidance, while the null-space term restores a preferred arm posture without affecting the Cartesian correction.

III. C-SPACE RMP FULL-BODY OBSTACLE AVOIDANCE

The geometric refinement of Section II moves the EEF path away from detected obstacles and projects the resulting displacement into joint space. This resolves straightforward EEF collisions efficiently, but it leaves a critical gap: the intermediate links of the manipulator are never explicitly monitored. A trajectory that is free at the EEF can still have the elbow, wrist, or upper arm penetrating an obstacle. To close this gap we introduce a Configuration-space RMP layer that reasons about every link simultaneously and generates coordinated joint corrections in real time. When the robot avoids an obstacle by applying a Cartesian correction to the EEF, the resulting joint motion is determined solely by the EEF Jacobian. This implicitly assumes that avoiding the EEF is sufficient for full-body safety, an assumption that breaks down whenever an obstacle is positioned near an intermediate link. Moving the EEF away from one obstacle can, in fact, swing the elbow *into* another. A principled solution must reason in the space of joint configurations rather than in Cartesian EEF space.

Riemannian Motion Policies (RMPs) provide exactly this framework. An RMP is a pair $(f, \mathbf{A})$ where f is a motion policy and $\mathbf{A}$ is a Riemannian metric that encodes the policy's

importance in each direction. In TRACE policies are defined as a per-link Cartesian spaces. RMPs *pull back* through the corresponding Jacobians into configuration space, where they are combined into a single consistent joint acceleration. Crucially, all links are treated simultaneously, so the combined policy cannot fix one link at the expense of another.

a) Approximate Link Kinematics: In the offline phase, the robot's full forward kinematics are not available for arbitrary candidate configurations. However, for collision reasoning, approximate link positions suffice. TRACE interpolates each link's position linearly between the shoulder origin and the EEF:

$$\vec{x}_k(t) \approx (1 - \alpha_k) \vec{p}_{\text{sh}} + \alpha_k \vec{x}_{\text{eef}}(t) + \delta_k, \quad (12)$$

where $\alpha_k \in [0, 1]$ scales the link's proximity to the EEF and δ_k is a curvature bias that accounts for elbow geometry. This provides sufficiently accurate link locations for collision detection at negligible computational cost.

b) Geometric Jacobian Mapping: To project link-space corrections into joint space, TRACE assembles the geometric Jacobian for each link k . For a revolute joint i , its contribution to the translational velocity of link k is the cross-product lever-arm formula:

$$\mathbf{J}_k^{(i)} = \hat{\mathbf{z}}_i \times (\vec{x}_k - \vec{x}_i), \quad i \leq k, \quad (13)$$

and zero for $i > k$, since distal joints do not affect proximal links. The full Jacobian for link k is then

$$\mathbf{J}_k = \begin{bmatrix} \mathbf{J}_k^{(1)} & \mathbf{J}_k^{(2)} & \dots & \mathbf{J}_k^{(7)} \end{bmatrix}. \quad (14)$$

This is the key structural difference from the scalar approximation $\mathbf{J}_k \approx \alpha_k \mathbf{I}$ used in EEF-only baselines: the true geometric Jacobian encodes how much each joint contributes to each link's velocity, correctly weighting joints by their lever-arm distance to the link they control.

c) Collision RMP Policy: For each link k and obstacle j , the signed clearance distance is

$$d_{kj} = \|\vec{x}_k - \mathbf{o}_j\| - r_j - r_k^{\text{col}}, \quad (15)$$

and the outward unit normal from the obstacle to the link is

$$\mathbf{n}_{kj} = \frac{\vec{x}_k - \mathbf{o}_j}{\|\vec{x}_k - \mathbf{o}_j\|}. \quad (16)$$

When the link enters the obstacle's influence region ($d_{kj} < r_{\text{inf}}$), a repulsive acceleration is generated:

$$\vec{f}_{kj} = (\eta_{\text{rep}} e^{-d_{kj}/\nu_{\text{rep}}} - \beta_{kj}) \mathbf{n}_{kj}. \quad (17)$$

The associated RMP metric is anisotropic:

$$\mathbf{A}_{kj} = \lambda_r \mathbf{n}_{kj} \mathbf{n}_{kj}^\top + \lambda_t (\mathbf{I} - \mathbf{n}_{kj} \mathbf{n}_{kj}^\top), \quad (18)$$

with $\lambda_r \gg \lambda_t$. This choice is deliberate: the metric strongly penalises motion *into* the obstacle (radial direction) while imposing little cost on motion *around* it (tangential directions). As a result, links naturally slide around obstacles rather than freezing or bouncing back sharply—behaviour that an isotropic metric cannot produce (6)

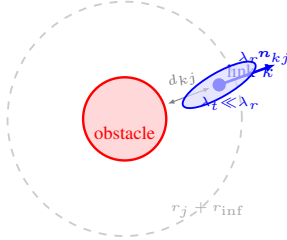


Fig. 6. Anisotropic metric ellipsoid at link k . The long axis (λ_r , radial) strongly opposes penetration; the short axis (λ_t , tangential) permits smooth sliding around the obstacle.

d) *C-Space Pullback and Joint Acceleration*: Each per-link, per-obstacle policy is pulled back into configuration space through the corresponding geometric Jacobian:

$$\mathbf{B}_{kj} = \mathbf{J}_k^\top \mathbf{A}_{kj} \mathbf{J}_k, \quad (19)$$

$$\mathbf{g}_{kj} = \mathbf{J}_k^\top \mathbf{A}_{kj} \vec{f}_{kj}. \quad (20)$$

Summing over all links and obstacles yields the global metric and generalised force:

$$\mathbf{B}_{\text{total}} = \sum_{k,j} \mathbf{B}_{kj}, \quad \mathbf{g}_{\text{total}} = \sum_{k,j} \mathbf{g}_{kj}. \quad (21)$$

The optimal joint acceleration is then obtained by a single linear solve:

$$\ddot{\mathbf{q}}^* = (\mathbf{B}_{\text{total}} + \varepsilon \mathbf{I})^{-1} \mathbf{g}_{\text{total}}. \quad (22)$$

The final joint displacement and its Cartesian EEF consequence are

$$\Delta \mathbf{q} = \gamma_{\text{gain}} \ddot{\mathbf{q}}^*, \quad \Delta \vec{x}_{\text{ef}} = \mathbf{J}_{\text{ef}} \Delta \mathbf{q}. \quad (23)$$

Using a calibrated gain γ_{gain} rather than strict Δt^2 integration prevents avoidance motions from becoming negligibly small at typical control timesteps.

e) *Temporal Smoothing*: Applying per-frame joint corrections without any filtering can excite high-frequency oscillations in the trajectory. TRACE smooths the Cartesian correction signal with a Gaussian kernel:

$$\widetilde{\Delta x}^{(d)}(t) = \int \Delta x^{(d)}(t') \frac{1}{\sqrt{2\pi}\sigma_T} e^{-(t-t')^2/2\sigma_T^2} dt'. \quad (24)$$

Additionally, a cosine ramp mask $m(t)$ suppresses the correction near the grasp and release phases of the task, where even small perturbations to the EEF pose could cause the robot to drop the object:

$$\Delta \vec{x}_{\text{ef}}^{\text{final}}(t) = m(t) \widetilde{\Delta \vec{x}_{\text{ef}}}(t). \quad (25)$$

IV. REACTIVE ONLINE EXECUTION

The offline refinement assumes fixed obstacle locations. During execution, however, obstacles may move or new collisions may emerge due to kinematic approximation errors. TRACE therefore re-solves the RMP system online at every control timestep using exact simulator Jacobians.

EEF Attractor. An isotropic Cartesian attractor tracks the reference trajectory:

$$\vec{f}_{\text{attr}} = k_p(\vec{x}^* - \vec{x}_{\text{ef}}) - k_d \dot{\vec{x}}_{\text{ef}}, \quad \mathbf{A}_{\text{attr}} = \frac{1}{2} \mathbf{I}. \quad (26)$$

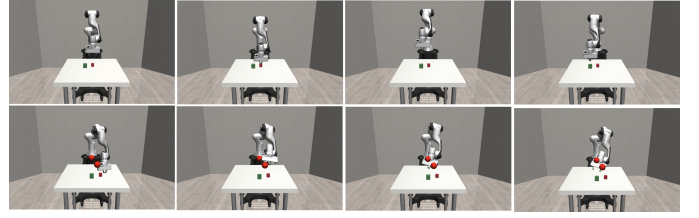


Fig. 7. Top: obstacle-free execution. Bottom: reactive online RMP correction avoids shifted obstacles while keeping the EEF close to the refined Gaussian-QP trajectory.

Online RMP Solve. The attractor, collision, and damping policies are combined as

$$\mathbf{B}_{\text{total}} = \mathbf{J}_{\text{ef}}^\top \mathbf{A}_{\text{attr}} \mathbf{J}_{\text{ef}} + \sum \mathbf{J}_k^\top \mathbf{A}_{kj} \mathbf{J}_k + \mu_d \mathbf{I}, \quad (27)$$

$$\mathbf{g}_{\text{total}} = \mathbf{J}_{\text{ef}}^\top \mathbf{A}_{\text{attr}} \vec{f}_{\text{attr}} + \sum_{k,j} \mathbf{J}_k^\top \mathbf{A}_{kj} \vec{f}_{kj} - \mu_v \dot{\mathbf{q}}. \quad (28)$$

The target joint velocity is updated through Euler integration:

$$\dot{\mathbf{q}}^*(t+1) = \dot{\mathbf{q}}(t) + \Delta t \ddot{\mathbf{q}}^*(t), \quad (29)$$

and the controller action becomes

$$\mathbf{a}_t = [\dot{\mathbf{q}}^*{}^\top, g_t]^\top. \quad (30)$$

This enables reactive full-body obstacle avoidance without offline re-planning.

V. EXPERIMENTS

Setup. TRACE is evaluated on the Franka Panda arm in MuJoCo. Demonstrations are collected using an OSC_POSE controller ($k_p = 150$) and transported to new scenes using [3]. Two spherical obstacles ($r_1 = 0.04$ m, $r_2 = 0.035$ m) are inserted at 40% and 55% of the trajectory arc length, causing the transported trajectory to collide with both obstacles (Figure 7).

Results. The geometric Jacobian formulation achieves full-body obstacle avoidance, whereas the scalar approximation $\mathbf{J}_k \approx \alpha_k \mathbf{I}$ leaves intermediate links in collision. The refinement stage preserves the transported trajectory structure with residual errors remaining near ~ 1 mm after deformation.

VI. CONCLUSION

We presented TRACE, a full-body collision-aware policy transportation framework combining offline geometric refinement with online C-space RMP avoidance. The Frenet-Serret and QP refinement stage generates smooth collision-free Cartesian trajectories, while the RMP layer provides coordinated full-body obstacle avoidance through geometric Jacobian pullback. The unified solve in Eq. (27) enables real-time reactive avoidance at 20 Hz. Future work includes exact offline FK, non-convex obstacle models, and learned deformation kernels.

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